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Activation of Lecithin Cholesterol Acyltransferase by a Disulfide-Linked Apolipoprotein A-I Dimer

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Apolipoprotein A-I_{Milano} is a molecular variant of apoA-I, containing the $Arg^{173} \rightarrow Cys$ substitution that forms a disulfide linked homodimer (A-I_M/A-I_M). To assess the effect of this structural modification on a major function of the apolipoprotein, its activation of lecithin cholesterol acyltransferase (LCAT), we prepared well-defined complexes of A-I_M/A-I_M and apoA-I with phospholipids and cholesterol and compared their reactivities with LCAT. Particles with A-I_M/A-I_M had very similar diameters to apoA-I particles (7.8 and 12.5 nm) but had distinct apolipoprotein and phospholipid contents and protein secondary structures; they bound LCAT with comparable affinities, but were less efficient substrates for the enzyme (40 to 70% less reactive). We conclude that the local structural constraints in A-I_M/A-I_M do not prevent the formation of well-defined complexes with phospholipids and do not influence the binding of the enzyme to the particles, but have an inhibitory effect on LCAT activation. © 1997 Academic Press

Apolipoprotein A-I (apoA-I), a single polypeptide chain composed of 243 amino acids, is the major protein constituent of high density lipoproteins (HDL). In addition to stabilizing HDL structure, apoA-I is the main activator of the enzyme lecithin cholesterol acyltransferase (LCAT), it plays a role in reverse cholesterol transport and it may be a ligand for HDL binding to cellular receptors (1-3). The entire sequence of the apoA-I gene has been determined; the gene is composed of 4 exons, the fourth one encodes a sequence with characteristic 22 amino acid repeats also found in other water-soluble apolipoproteins (4). These repeats are believed to form repetitive antiparallel amphipathic α-helices that are thought to be responsible for the main biological activities of apoA-I. Reconstituted HDL (rHDL) have provided very useful models to investigate the structure and function of apoA-I in well-defined lipid complexes (5). A number of studies have shown that distinct structures of apoA-I may exist in different rHDL particles, which correlate with their reactivity with LCAT (6,7).

Apolipoprotein A-I_{Milano} (A-I_M) is a molecular variant of apoA-I, characterized by the $Arg^{173} \rightarrow Cys$ substitution (8), that forms a disulfide-linked homodimer (A-I_M/A-I_M). The introduction of the disulfide bridge in A-I_M/A-I_M results, for the lipid-free protein, in enhanced interhelix interactions, a low tendency to self associate, increased secondary structure, and a more folded tertiary structure (9). In this study, we used defined complexes with phospholipids to compare the LCAT activation by A-I_M/A-I_M and apoA-I in order to evaluate the impact of the interchain disulfide bridge in A-I_M/A-I_M on its functional properties.

MATERIALS AND METHODS

A-I_M/A-I_M, expressed in *E. coli* and purified by conventional chromatographic procedures (9), was a generous gift of Pharmacia and Upjohn, Stockholm, Sweden. ApoA-I and LCAT were purified from human plasma, as previously described (10,11). The purity of the proteins (> 95%) was confirmed by sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) using Coomassie Blue and silver staining. Before use, lyophilized A-I_M/A-I_M and apoA-I were dissolved in 20 mM phosphate buffer, pH 7.4, containing 6 M GdnHCl and extensively dialyzed against 10 mM Tris-HCl, pH 8.0, 150 mM NaCl, 1 mM NaN_3, and 0.01% EDTA (reconstitution buffer). Low density lipoproteins (LDL) (d = 1.020-1.050 g/ml) were isolated and washed at their upper density by ultracentrifugation.

Preparation and characterization of rHDL. Discoidal rHDL containing A-I_M/A-I_M or apoA-I were prepared by the cholate dialysis technique (12), using sodium cholate: palmitoyloleoyl phosphatidylcholine (POPC) at a molar ratio of 1:1 and 5 mg of apolipoprotein in each preparation. Two initial POPC:cholesterol:apolipoprotein weight ratios, 2.17:0.11:1 and 3.26:0.08:1, were used for both A-I_M/A-I_M and apoA-I. Since the investigated apolipoproteins differ in molecular weight, weight rather than molar ratios were used to prepare comparable rHDL particles. These weight ratios correspond to POPC:cholesterol:apoA-I molar ratios of 80:8:1 and 120:6:1. After removal of sodium cholate by extensive dialysis against reconstitution buffer, the 9.6 and 12.7 nm apoA-I rHDL, as well as the 12.5 nm A-I_M/A-I_M rHDL, were isolated from the reaction mixtures containing

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3.26:0.08:1, POPC:cholesterol: apolipoprotein, by gel filtration on three serial Superose-6 columns. The small 7.8 nm rHDL, containing apoA-I or A-I_M/A-I_M, were prepared from the reaction mixtures containing 2.17:0.11:1, POPC:cholesterol: apolipoprotein, after incubation with LDL for 18 h at 37°C (6,13).

Sizes and size distributions of the rHDL complexes were obtained by nondenaturing gradient gel electrophoresis (GGE) (14) on precast 8-25% polyacrylamide gels (Pharmacia Biotec), using the Pharmacia Phast System. After Coomassie Blue staining, the gels were scanned with a Pharmacia Ultroscan XL laser densitometer. The size of rHDL particles was calculated with the Pharmacia 2400 Gelscan XL software, using thyroglobulin (17.0 nm), apoferritin (12.2 nm), catalase (9.2 nm), lactate dehydrogenase (8.2 nm), and bovine serum albumin (7.1 nm) as calibration proteins.

The number of apolipoprotein molecules per particle was determined by cross-linking with dimethylsuberimidate (DMS) (15). Cross-linked samples were analyzed by SDS-PAGE on 4 to 10% polyacrylamide gradient slab gels, using the Tris-tricine buffer system of Schagger and Van Jagow (16) in a Mini Protean slab minigel apparatus (BioRad). The chemical composition of the rHDL was determined by standard methods as reported previously (6,7).

Circular dichroism (CD) spectra were recorded with a Jasco J500A spectropolarimeter at a constant temperature of 25°C. The α-helical content was calculated from the empirical expression of Chen et al. (17): $[\theta]_{222} = -30{,}300 f_H - 2{,}340$.

LCAT reaction kinetics. The rHDL complexes containing A-I_M/A-I_M or apoA-I were preincubated with aliquots of [^{14}C]cholesterol ($\approx 4 \times 10^4$ cpm) dispersed in 2% defatted bovine serum albumin at 37°C for 1 h to label the substrate particles (6). Reaction mixtures for kinetic analysis had substrate concentration ranging from 1.8×10^{-7} M to 10^{-5} M expressed in terms of A-I_M and apoA-I monomers, 2 mg of defatted bovine serum albumin and 5 to 125 ng of pure LCAT in 0.5 ml of the recombinant buffer. In most experiments, 4 mM β-mercaptoethanol was included in the mixture. The reaction time varied from 30 min to 1 h, depending on the reactivity of the complex; the lipids were extracted with 5 ml of chloroform/methanol, labeled CE were separated from FC and PL by thin layer chromatography and were quantified by scintillation counting. In general, the percentage of [^{14}C]cholesterol converted to cholesterol ester was less than 10%, ensuring that initial velocity values were being measured. Determination of the apparent kinetic constants, apparent V_{max} and apparent K_m was performed using Lineweaver-Burk plots of the kinetic data.

Activity inhibition assay for determining the binding of LCAT to rHDL. The standard substrate was eggPC:cholesterol:apoA-I rHDL prepared in 100:5:1 molar ratios and including [^{3}H]DPPC. The LCAT reaction mixtures contained standard substrate, 2 mg of defatted BSA, unlabeled test rHDL, β-mercaptoethanol, and standard buffer to 0.45 ml total volume. Tubes were incubated at 37°C in a shaking water bath for 15 min, and 12.5 ng of pure LCAT in 50 μl were added to start the reaction. Analysis of the results to obtain LCAT inhibition (i.e., dissociation) constants with the unlabeled test rHDL was performed as described previously (18).

RESULTS

Preparation and characterization of A-I_M/A-I_M rHDL. The chemical composition and size of the isolated rHDL containing A-I_M/A-I_M or apoA-I are shown in Table I. Chemical cross-linking of A-I_M/A-I_M in rHDL and analysis of the products by SDS-PAGE indicated that the two rHDL with diameters of 7.8 and 12.5 nm contained 1 and 2 A-I_M/A-I_M molecules/particle, respectively. As previously described 7.8, 9.6, and 12.7 nm apoA-I rHDL contained 2, 2, and 3 apoA-I molecules/particle, respectively (19). CD measurements show that the 9.6 and 12.7 nm apoA-I rHDL display the highest α-helical content. Small and large A-I_M/A-I_M rHDL had similar α-helical contents ($\sim$ 60%), comparable to that of the 7.8 nm apoA-I rHDL.

LCAT activation. The reactivity of apoA-I and A-I_M/A-I_M rHDL particles of comparable size with LCAT was determined by measuring the initial velocity of the LCAT reaction as a function of apoA-I and A-I_M/A-I_M concentration in rHDL. The 9.6 apoA-I rHDL particles were included as a well characterized, and highly reactive substrate control for LCAT (7). Table II and Figure 1 summarize the kinetic parameters for the reaction of LCAT with the five particles. Consistent with previous results (6,7) the 7.8 nm rHDL are poor substrates for LCAT, compared with the 12.5-12.7 nm rHDL, which display a reactivity similar to that of 9.6 nm apoA-I rHDL. Using the apparent V_{max}/apparent K_m parameter as the measure of the substrate efficiency, the 12.7 nm A-I_M/A-I_M rHDL are 70% less reactive with LCAT than their apoA-I size homologues (Table II). This lower reactivity is mainly due to differences in apparent K_m, which may reflect the binding affinity of the enzyme to the rHDL (18); the apparent V_{max} values, which indicate the activation of the enzyme and the catalytic rate constants (18), are essentially the same for the two particles. A similar trend was observed for the 7.8 nm rHDL, with A-I_M/A-I_M rHDL being 40% less reactive than apoA-I rHDL (Table II). When the apparent V_{max}/apparent K_m parameters are expressed in terms of phospholipid concentrations, the differences in reactivity between the A-I_M/A-I_M and apoA-I homologues become smaller (Figure 1).

To determine whether the large differences in the apparent K_m values are due to major differences in the affinity of the enzyme for the substrate particles, we determined the dissociation constants (K_d) for all the particles, using the affinity-inhibition method developed by Bolin and Jonas (18). Figure 2 shows typical Lineweaver-Burk plots of the reciprocal of initial velocity versus the reciprocal of standard substrate concentration in the presence of increasing concentrations of 7.8 A-I_M/A-I_M rHDL. Then a plot (not shown) of the slopes as a function of "inhibitor" particle concentration gives the K_d for LCAT binding to the A-I_M/A-I_M rHDL (18). Table III lists the K_d values determined by this method in terms of apolipoprotein monomer concentration, particle concentration and PC concentration. In all cases the differences in K_d values were at most 4.7-fold. These results indicate that the protein content and conformation in the particles does not have a major effect on the binding of LCAT and does not account for the differences in reactivity between the 7.8 nm and larger particles. This is in agreement with our previous results showing that binding of LCAT to rHDL depends mostly on the phospholipid composition rather than on the apolipoprotein component (18).

TABLE I

Properties of Reconstituted HDL

Particle	Size[a] (nm)	POPC/Chol/Protein[b] (w/w/w)	Protein/rHDL[c]	α-helix[d] (%)
7.8 mm apoA-I rHDL	7.8	1.13(±0.04)/0.09(±0.02)/1	2	59
9.6 nm apoA-I rHDL	9.6	2.05(±0.07)/0.07(±0.02)/1	2	71
12.7 nm apoA-I rHDL	12.7	3.20(±0.05)/0.09(±0.02)/1	3	71
7.8 nm A-I_M/A-I_M rHDL	7.8	0.82(±0.15)/0.06(±0.02)/1	1	60
12.5 nm A-I_M/A-I_M rHDL	12.5	1.92(±0.08)/0.15(±0.02)/1	2	59

[a] Diameter determined by nondenaturing polyacrylamide gradient gel electrophoresis (±0.1 nm, SD).
[b] Determined from three independent reconstitution experiments (± SD).
[c] Determined by SDS-PAGE of delipidated particles after cross-linking with DMS.
[d] Calculated from the molar ellipticity at 222 nm using the empirical expression of Chen et al. (17).

Since the kinetic experiments were performed in the presence of β-mercaptoethanol, which is known to increase the activity of LCAT (1,2), we wanted to assess the effects of this reducing agent on the reactivity of the particles containing A-I_M/A-I_M dimers. Kinetic experiments performed in the absence of β-mercaptoethanol showed an equivalent 3- to 4-fold decrease in apparent V_{max} and app V_{max}/app K_m both for the apoA-I and the A-I_M/A-I_M containing rHDL particles (results not shown). This observation indicates that under our experimental conditions, β-mercaptoethanol only affects LCAT activity and not the properties of the particles.

DISCUSSION

There is strong evidence that apoA-I, in different complexes with lipids, can activate LCAT with very different efficiency (2,6,7). Indeed, a number of studies have identified a correlation between rHDL particle size and their ability to interact with LCAT (6,7). Therefore, in studies of apoA-I variant structure and function, it is imperative to compare complexes of apoA-I and the variant with lipids which are identical in size. Here we prepared and isolated two sets of rHDL particles with either apoA-I (7.8 nm and 12.7 nm) or A-I_M/A-I_M (7.8 nm and 12.5 nm). As expected, the larger particles were more effective substrates for LCAT than the smaller ones, independent of protein composition. Within particles of comparable size, those containing A-I_M/A-I_M were 40-70% less reactive than those with apoA-I.

In view of the large differences in apparent K_m values we measured in this study, we set out to determine whether the binding of LCAT to the rHDL is affected by the presence of the disulfide-linked apoA-I or by particle diameter. The K_d values measured by the activity-inhibition method (18) do not correlate with the rHDL protein composition nor account for the large differences in the reactivity of the 12.5 and 12.7 nm rHDL on the one hand and the 7.8 nm rHDL on the other. The absence of clear trends and major differences in K_d with these particles agrees with our previ-

TABLE II

Kinetic Constants for the Reaction of LCAT with ApoA-I and A-I_M/A-I_M rHDL Particles

Particle	Apparent V_{max} (nmol CE/hr.)	Apparent K_m (M) ($\times 10^7$)	App. V_{max} / App. K_m (nmol CE/hr. M) ($\times 10^{-6}$)
9.6 nm apoA-I rHDL[a]	0.226 (±0.016)	1.37 (±0.27)	1.68 (±0.22)
12.7 nm apoA-I rHDL[a]	0.296 (±0.019)	1.67 (±0.07)	1.7 (±0.13)
12.5 nm A-I_M/A-I_M rHDL[a]	0.373 (±0.033)	7.48 (±1.02)	0.500 (±0.027)
9.6 nm apoA-I rHDL[b]	0.900	6.98	1.29
7.8 nm apoA-I rHDL[b]	0.244	70.8	0.035
7.8 nm A-I_M/A-I_m rHDL[b]	0.537	268	0.020

Analysis of kinetic results by Lineweaver-Burk plots gave straight lines which had correlation coefficients from 0.986 to 0.999.
[a] Experiments were performed in triplicate using substrate concentrations from 2.43×10^{-7} to 2.43×10^{-6} M expressed in terms of apoA-I and apo A-I_M monomers. The kinetic parameters are the means (± standard deviations).
[b] A separate set of experiments was performed with distinct particle and LCAT preparations. The results are the averages of duplicate measurements using substrate concentrations from 1.86×10^{-7} to 9.75×10^{-6} M.

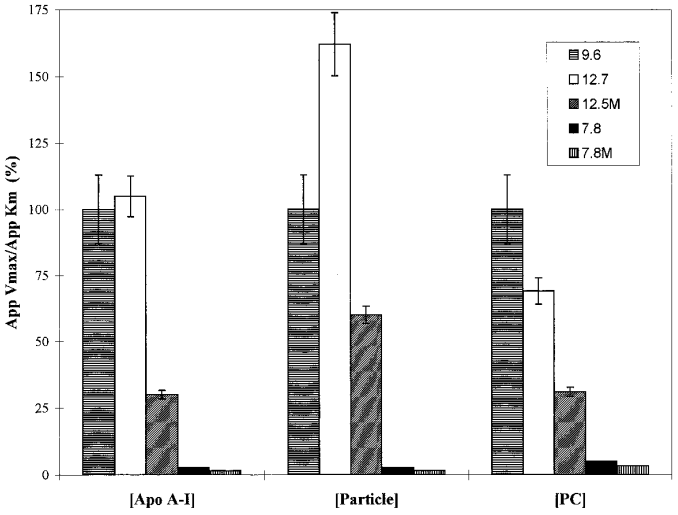


FIG. 1. Relative reactivity of rHDL particles with LCAT. The % apparent V_{max}/apparent K_m is shown for the apoA-I rHDL and A-I_M/A-I_M rHDL particles relative to the 9.6 nm apoA-I rHDL. The apparent V_{max}/K_m parameters were calculated as a function of apoA-I monomer concentration, particle concentration, and phospholipid (POPC) concentration. In each of these three sets of data, the particles are from left to right: 9.6 nm apoA-I rHDL; 12.7 nm apoA-I rHDL; 12.5 nm A-I_M/A-I_M rHDL; 7.8 nm apoA-I rHDL; 7.8 nm A-I_M/A-I_M rHDL.

ous findings that the binding of LCAT to the surface of rHDL is mostly determined by the lipid composition of the particles (18). LCAT kinetics are usually described using the model of Verger et al. (20) to interpret kinetics of enzymatic catalysis at interfaces. This model does not take into account the fact that LCAT binding to the interface and its activation are separate

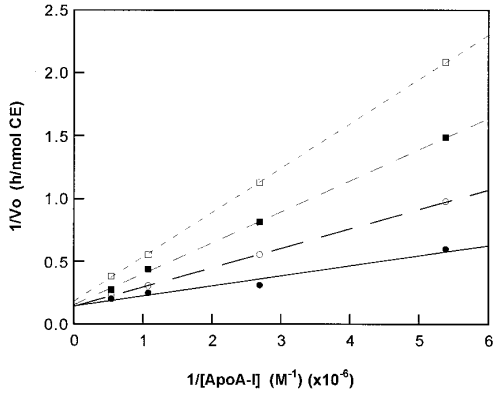


FIG. 2. A typical kinetic experiment for the determination of K_d by the activity-inhibition method (18). The reciprocal of the initial velocity is plotted versus the reciprocal of the standard substrate concentration (rHDL labeled with ^{3}H-DPPC) in the presence of increasing concentrations of the "inhibitor" particle 7.8 nm A-I_M/A-I_M rHDL: (●) no "inhibitor"; (○) 5.6×10^{-7} M; (■) 1.4×10^{-6} M; (□) 2.8×10^{-6} M. The "inhibitor" concentration was subsequently plotted versus the slopes of these Lineweaver-Burk plots to obtain the inhibition constant, K_d.

TABLE III

Dissociation Constants (K_d) of LCAT from ApoA-I and A-I_M/A-I_M rHDL Particles Determined by the Activity-Inhibition Methods

Particle	K_d (M) [Protein] ($\times 10^7$)	[Particle] ($\times 10^7$)	[PC] ($\times 10^5$)
7.8 nm apoA-I rHDL	4.32	2.16	1.77
9.6 nm apoA-I rHDL	2.11	1.06	1.62
12.7 nm apoA-I rHDL	3.83	1.28	4.48
7.8 nm A-I_M/A-I_M rHDL	9.87	4.94	3.55
12.5 nm A-I_M/A-I_M rHDL	2.49	0.63	1.84

The K_d values are expressed as a function of apolipoprotein monomer concentration, particle concentration, and phospholipid concentration. All measurements were performed in duplicate. The linear regression analysis of the series of Lineweaver-Burk plots ($1/v_0$ versus 1/[S]) gave correlation coefficients from 0.989 to 0.999, and the plots of the slopes versus inhibitor concentration gave correlation coefficients from 0.989 to 1.000.

steps. Furthermore, the model lacks the multiple catalytic steps found in the LCAT reaction (21). However, for the sake of argument, and in the absence of a better model, we can assume that the experimental apparent V_{max} and apparent K_m are represented by the simplified equations:

$$\text{app.}V_{max} = \frac{k_{cat}E_0 \cdot S}{S + K_m^*}\ ;\ \text{app.}K_m = K_d\,\frac{K_m^* \cdot S}{S + K_m^*}$$

The major differences we observed in the apparent K_m values between the A-I_M/A-I_M rHDL and their apoA-I size homologues should then be attributed to the intrinsic K_m^* values, as the measured K_d values are not markedly different, and the interfacial concentration of the substrates S (phospholipid and cholesterol), in molecules per unit surface, are comparable. The measured apparent V_{max} for the various particles are not dramatically different; thus, changes in k_{cat} would have to compensate for changes in K_m^*. Changes of this nature in the intrinsic constants would require conformational changes in the the active site of LCAT which alter not only the binding of the lipid substrates but also the catalytic rate constant. Most likely, such conformational differences in LCAT would be related to differences in the activation of the enzyme by different apolipoprotein structures in the various particles (22).

The central region of apoA-I (approximately residues 99-186) clearly plays a key role in LCAT activation. Within the central region, the strongest data implicate a domain comprising a couple of amphipathic helices which can adopt different conformations depending on rHDL size (22). Several reports based on deletion mutagenesis indicate that any deletion in apoA-I which dis-

rupts the conformation of the central domain results in reduced LCAT activity (23-25), providing the final evidence that a critical conformation rather than sequence is required for LCAT activation. Computer modeling of A-I_M/A-I_M in rHDL shows that the helical segments from the disulfide bridge to the N-terminus of each A-I_M monomer can form the expected cylindrical shell of discoidal rHDL but with a somewhat distorted structure and reduced inner diameter (Calabresi et al., 1996; unpublished results). Therefore, the relative decrease in reactivity of A-I_M/A-I_M rHDL compared to apoA-I rHDL is most likely due to the distortion of the local structure near the disulfide bond, but the overall structure remains competent in LCAT activation.

In conclusion, the constraints imposed by the disulfide bridge in A-I_M/A-I_M result in a reduced ability to activate LCAT but do not have a significant effect on the binding of the enzyme to the surface of the rHDL particles.

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